Gross Margin Decline, Q1 → Q3 2026

Gross margin fell from 33.3% (Q1) to 19.6% (Q3): gross profit down ₦977M, from ₦1,727M to ₦750M (–57%). 59% of that loss (₦572M) came from six Distributor customers.

From 1 July these six buy only sweeteners (100% of their kilos, against 18.6% in Q1; 3 of the 20 products), and sweeteners earn them a 3.1% margin, down from 33.3%.

The other ₦405M: six Bakery/Confectionery customers on 10–13% discounts (–₦193M) and all remaining customers (–₦212M, mainly FX raising USD costs by ₦347M while NGN price increases recovered only ₦186M).

Q1 and Q3 reconcile to Finance's control totals (Q1 ₦5,191.2M at 33.3%; Q3 ₦3,825.6M at 19.6%); 13 of 13 validation checks pass.

Recommendation: tighten discount approval, set a margin floor on distributor sweeteners, review prices monthly against FX, and find out *why* the distributors changed before committing to a win-back plan (§2).

v1.0 · Author: Ogunlusi Teniola Elizabeth, Data Analyst · Date: 17 October 2026 · Database: SQLite · Run in DBeaver

---

Contents

1. Findings · 2. Decision and recommendation · 3. Definitions and method · 4. Data cleaning log · 5. Reconciliation and validation · 6. How to rerun · 7. Limitations and confidence · 8. Open items · 9. Files

---

1. Findings

All figures come from files 01 to 05 unless marked otherwise (§2 marks the exceptions). Amounts are naira millions (₦M).

1.1 Where the ₦977M went (gross profit, Q3 vs Q1) — `04_margin_bridge.sql`, query 4a

Group	GM Q1	GM	Change	Share of

		Q3		drop
Distributors (C009, C018, C027, C036, C045, C054)	602.2	30.0	**–572.2**	**58.6%**
Deep-discount customers (C007, C019, C033, C061, C062, C063)	334.7	141.5	–193.2	19.8%
All other customers	790.0	578.2	–211.8	21.7%
Total	**1,726.9**	**749.7**	**–977.2**	100%

A *deep-discount customer* is any customer with at least one line discounted 10% or more: four Bakery and two Confectionery customers.

1.2 Why, by driver — query 4c (each column sums exactly to its change)

Driver	Distributors	Deep-discount	All other	Total
Volume / mix	–514.5	–89.2	–112.7	**–716.4**
FX on USD supplier costs	–106.3	–64.9	–175.9	**–347.0**
NGN list-price increases	+46.2	+37.5	+102.2	**+185.8**
Discounts	+2.4	–69.8	–5.6	**–73.1**
Supplier cost increases (P03 and P04, 1 June)	0.0	–6.7	–19.8	**–26.5**
Change in gross profit	–572.2	–193.2	–211.8	**–977.2**

Net of price increases, FX cost ₦161M (–347.0 + 185.8). Volume/mix is dominated by the distributor product-mix collapse (–514.5).

1.3 Three findings

1. **Distributor behaviour change (₦572M).** Query 4d: Q1 had 47 orders, 223k kg and 33.3% margin; Q3 has 175 orders (3.7×), 682k kg and 3.1% margin, with sweeteners 100% of kilos. In `margin_model` the distributors bought 20 distinct products in Q1 and 3 in Q3. The three sweeteners are Maltodextrin (P06), Sorbitol 70% (P07) and Dextrose Monohydrate (P08). *The cause of the change is unknown (§7).*

2. **FX is outrunning price.** Of the ₦347M FX cost increase, NGN list-price increases recovered ₦186M. The naira moved from ₦1,480 to ₦1,720 per USD between January and September (query 4b).
3. **Deep discounts began in July.** 350 lines at 10–13%, none before 1 July. The six customers' average discount rose from 3.1% to 11.6% between Q1 and Q3 (query 4c shows discounts cost ₦69.8M of margin). Whether the discounts retained volume is addressed in the supporting figures (§2).

1.4 When the margin slipped — query 4b

Month	Net rev ₦M	GM %	GM % ex-distributors	GM % distributors	Discount %	FX ₦/$
Jan	1,760.0	33.4	33.3	33.5	2.2	1,480
Feb	1,804.0	33.8	33.6	34.1	1.7	1,495
Mar	1,627.2	32.5	32.8	31.7	2.4	1,520
Apr	1,643.3	31.4	31.8	30.5	2.0	1,540
May	1,669.6	31.7	31.5	32.2	1.9	1,580
Jun	1,550.2	29.0	28.9	29.5	2.2	1,620
Jul	1,404.4	**21.5**	26.6	**4.6**	4.0	1,660
Aug	1,241.7	19.0	24.5	2.8	4.3	1,690
Sep	1,179.5	17.9	23.8	1.9	3.9	1,720

- **Jan–Feb:** stable at 33.3–33.6% (ex-distributors).
- **Mar–May:** a steady drift to 31.5%: FX moving, prices not.
- **Jun:** a step to 28.9%: supplier cost increases plus another FX move.
- **Jul:** the structural break: distributors switch to sweeteners and deep discounts begin.

Is Q1 a fair reference? As a budget and board baseline, yes. As a run-rate, no: margin was already easing in March, and Q1 FX (₦1,480–1,520) no longer exists. Jan–Feb is the cleaner "undisturbed" baseline.

---

2. Decision supported and recommendation

Decision: Commercial committee — what to change in the price list, the discount rules, and which products and segments to push.

#	Recommendation	Why (evidence)	Indicative impact per quarter*	Main risk
B	**Cap discounts at 5% for the six deep-discount accounts, unless approved by name**	Discounts cost ₦69.8M of margin (4c)	+₦54M at a 5% cap if volume holds (+₦30M at 8%); pays off unless they lose more than ~28% of volume	Losing the accounts
C	**Set a minimum margin on distributor sweeteners (8% suggested: +5.3% on price)**	3.1% margin and falling with each FX move (4d)	+₦50M if volume holds; if they leave, margin lost is ₦30M	Gross margin excludes freight and warehousing; 682k kg may be loss-making in reality
A	**Raise non-distributor prices +5% to +8%, then review monthly against FX**	FX cost ₦347M vs price recovery ₦186M (4c)	+5%: +₦57M to +₦144M; +8%: +₦135M to +₦230M (volume loss 10% → 0%)	Price elasticity unknown
D	**Find out why distributors stopped buying 17 products, before committing to win-back**	Distributor volume/mix −₦514.5M (4c)	Their Q1 margin on these lines was ₦597M/quarter; recovering a quarter ≈ ₦150M (*scenario, not forecast*)	Root cause unknown

A + B + C ≈ ₦150–250M per quarter, roughly 15–25% of the ₦977M gap (overlapping, so not exactly additive). Closing the remainder depends on D. The FX table ends in September: a further naira move pushes distributor sweeteners below cost.

***Supporting figures calculated outside files 01 to 05.** The impact estimates, and these figures, were calculated separately from the same cleaned data and are **not** produced by the SQL files. Treat them as indicative until a script reproduces them:

- For non-distributor customers, like-for-like unit COGS rose 14.1% against a 4.9% rise in NGN prices; restoring Q1 margin on that book at September FX would need about +12.3% on price.

- The six deep-discount customers' volume fell 26.4% between Q1 and Q3; all other non-distributor customers' volume fell 3.4%. This suggests the discounts did not retain volume, but causality cannot be established (they may have been discounted because they were at risk).
- Option sizing is at the Q3 run-rate, in isolation, assuming volume holds or falls 5–10%.

---

3. Definitions and method

Finance definition (confirmed with Finance):

- `GM % = (Net revenue − COGS) ÷ Net revenue`
- **Net revenue** = kg × unit price (NGN) × (1 – discount%), after discounts and excluding VAT. USD-tagged lines are converted at that month's rate.
- **COGS** = kg × supplier cost (USD/kg) in effect **on the order date** × that **month's** FX rate. Freight, duty and warehousing are **excluded**.
- **Periods** by order date: Q1 = 1 Jan–31 Mar 2026; Q2 = Apr–Jun; Q3 = 1 Jul–30 Sep.
- **Groups:** Distributor = `customers.segment = 'Distributor'` ; deep-discount = not a distributor, and at least one line with a discount of 10% or more; all other = the rest.

Bridge method (04c). Per product within each group, volume-weighted, with Q1 as baseline (P = gross NGN price/kg, d = discount rate, u = USD cost/kg, f = FX, V = kg):

Effect	Formula
NGN list price	$(P_1 – P_0)(1 – d_0) \cdot V_1$
Discounts	$–(d_1 – d_0) \cdot P_1 \cdot V_1$
Supplier cost	$–(u_1 – u_0) \cdot f_0 \cdot V_1$
FX	$–(f_1 – f_0) \cdot u_1 \cdot V_1$
Volume / mix	$(V_1 – V_0)$ · baseline unit margin

The five effects sum exactly to the change in gross profit (check V12). The attribution is sequential: rate effects are measured at Q3 volumes and volume/mix at Q1 unit margins, so interaction terms sit inside the rate effects. A different ordering would shift amounts between rows but not the totals.

---

4. Data cleaning log

Raw: six tables (`orders` , `order_lines` , `products` , `customers` , `supplier_costs` , `fx_rates`). The source xlsx and CSVs have identical contents. Profiling (`01_profile.sql`) found no orphan keys.

Issue	Amount	Treatment	Basis
`order_lines` loaded twice	9,612 raw rows = 2 × 4,806; 4,806 copies removed	Removed by the same de-duplication rule as below	Load defect in my environment; see `01_profile.sql`
Exact duplicate lines (sync bug)	**94 lines; ₦248.6M of net revenue removed**	Keep the lowest `line_id` ; remove the other copies	IT-confirmed sync bug; Finance rule: exact copies only (same order, product, qty, price, discount, currency; only `line_id` differs)
Mixed date formats	177 orders in dd/mm/yyyy	Parsed day-first	Proof: 88 have day > 12, **0** have month > 12
Blank currency	107 lines	Treated as NGN	Finance: field added later, never mandatory
USD-tagged lines	27 lines	Converted at that month's FX	Finance: genuinely USD-priced
Rep names	4 people, 6 spellings; 88 orders blank	Standardised; blank = "Unassigned"	My decision; no effect on margin totals
Bag-product quantities (P03 Potassium Sorbate, P16 Sodium Bicarbonate)	239 lines on or after 1 May	kg = qty × 25	Proof below; Finance: "prove it from the data"

After cleaning: 4,712 lines and 1,890 orders.

Bag-unit proof (`05_validation_checks.sql` , 5b). Before 1 May the bag products' quantities are all multiples of 25 (25 to 3,000 kg). From 2 May they are 1, 2, 4, 8, 10, 20, 40, 80 and 120, none a multiple of 25, with zero overlap (last kg-style line 28 Apr; first bag-style line 2 May). Without the rule, Q3 net revenue is ₦3,565.3M, which is 6.8% short of Finance's ₦3,825.6M; with it, it ties exactly.

---

5. Reconciliation and validation

Period	Net revenue ₦M	Finance ₦M	Diff	GM %	Finance GM %	Result
Q1	5,191.2	5,191.2	0.0%	33.27	33.3	✅
Q3	3,825.6	3,825.6	0.0%	19.60	19.6	✅

Tolerance agreed with Finance: ±0.5% revenue, ±0.2 pts GM. **Q2 has no Finance control total**, so April to June come from the same validated pipeline but are not independently reconciled.

`05_validation_checks.sql` (query 5a) returns 13 checks, each PASS or FAIL, and all must pass:

Group	Checks
Tie-out to Finance	V01 to V04: Q1 and Q3 revenue and GM%
Row integrity	V05 model rows = clean lines (4,712); V06 one row per `line_id` ; V07 no line lacks a cost or FX rate; V10 no clean line lacks an order
Cleaning	V08 no invalid dates; V09 no exact duplicates left; V13 94 sync-bug duplicates removed
Rules and maths	V11 bag-unit break is clean (0 ambiguous quantities); V12 bridge total = actual change in gross profit

Verification record.

Run	Result
	Files 02 to 05 ran without error

My run in DBeaver (SQLite): 17 Oct 2026	

---

6. How to rerun

```
margin_analysis/
├── README.md
├── sql/
│   ├── 01_profile.sql             profile the raw data (read-only)
│   ├── 02_clean.sql               clean_order_lines, clean_orders
│   ├── 03_margin_model.sql        margin_model (one row per clean line, all mone
│   ├── 04_margin_bridge.sql       who, when, why (bridge_prod table + summaries)
│   └── 05_validation_checks.sql   13 PASS/FAIL checks, evidence, cleaning log
└── data/raw/                      the six CSVs from distributor_day1_csvs_2.zip
```

1. Load the six CSVs into tables named `orders`, `order_lines`, `products`, `customers`, `supplier_costs`, `fx_rates`. **Load each table exactly once.**
2. Run the files **in order, 01 to 05**, one script each (DBeaver: **Alt+X** runs a whole script; **Ctrl+Enter** runs one statement).
3. Compare each result with its `EXPECT` comment. The last check is the 13-row summary in 05, where every row must say **PASS**.
4. Each of files 02, 03 and 04 drops and recreates its own tables, so any file can be rerun safely. Raw tables are never modified.

Dialect notes (the scripts are written for SQLite). On PostgreSQL or SQL Server: dates need real date types (`date()` and text comparisons of dates are SQLite habits); `SUM(condition)` becomes `SUM(CASE WHEN condition THEN 1 ELSE 0 END)`; a subquery in `FROM` needs an alias; `LIMIT 1` becomes `TOP 1` (SQL Server); `CREATE TABLE … AS SELECT` becomes `SELECT … INTO` (SQL Server). Keep the `1.0*` in 04c on every database (integer division).

---

7. Limitations and confidence

Claim	Confidence	Why not higher
Reconciliation, cleaning, bag-unit rule, FX/price/discount figures	~95%	Q1 and Q3 tie to Finance exactly

The distributor shift is real behaviour, not a second data bug	~85%	Judged from the order pattern; at least 5 of the 6 distributors ordered in every month (all 6 in Jul–Sep); 26 near-duplicate distributor lines in Q3 are unexplained
FX lag explains most of the non-distributor slide; deep discounts did not buy volume	~80%	Observational; discounts may have been defensive
Why distributors changed behaviour	**<50%**	Not in the data

Limitations:

1. **Causality.** This analysis shows what changed and when, not why. Price elasticity cannot be estimated from this data, so the option sizing assumes volume holds or falls 5–10%.
2. **Gross margin excludes freight, duty and warehousing** (Finance's definition). Low-margin, high-volume sweeteners may be loss-making on a contribution basis.
3. **The bag-unit change date (1 May) is inferred from the data**, not confirmed by the warehouse.
4. **FX is monthly.** The month's rate applies to every order in that month, and the table ends in September.
5. **Blank currency = NGN** relies on Finance's statement; the prices look NGN-scale, which is consistent with it.
6. **VAT.** The data has no VAT field. Reconciling to Finance's ex-VAT figures to the decimal implies prices are already ex-VAT.
7. **Exact duplicates only.** Lines that share an order and product but are not exact copies are kept, per Finance's rule.
8. **Profiling cannot see meaning.** My table-level checks missed the bag-unit change; it surfaced only when Q3 failed to reconcile.
9. **Value checks are hard-coded for this extract** (Finance totals, the 94 duplicates). A refreshed extract needs new expectations.
10. **Groups are rule-defined** (segment label; any line ≥10% discount), and Q2 is not externally reconciled.
11. **Supporting figures in §2** are not produced by the SQL files.

---

8. Open items to close before the committee

1. Sales ops: is there a July contract, promotion or pricing agreement with the six distributors on sweeteners?
2. Account owners: did the distributors move their other 17 products to a competitor?
3. Commercial: who approved the 10–13% discounts, and on what evidence (competitor quote, churn risk)?
4. Finance/logistics: freight and warehousing ₦/kg, to test whether distributor sweeteners are loss-making.
5. Warehouse: confirm that bag products were recorded in bags from 1 May.
6. IT: confirm whether the surge in distributor orders (47 to 175) is genuine.

---

9. Files

File	Purpose
`sql/01_profile.sql`	Read-only profiling of the raw data (checks 1a to 1f)
`sql/02_clean.sql`	Builds `clean_order_lines` (4,712 lines) and `clean_orders` (1,890 orders)
`sql/03_margin_model.sql`	Builds `margin_model` : one row per clean line, all money in naira
`sql/04_margin_bridge.sql`	Who (4a), when (4b), why (4c bridge), distributor detail (4d)
`sql/05_validation_checks.sql`	13 PASS/FAIL checks, bag-unit evidence, cleaning log
`README.md`	This document